"The API is the bridge between your UI and your business logic."

MODULE 35

Frontend to API Integration

Building seamless connections between your React UI and the Spring Boot backend -- Axios, the Fetch API, SOAP consumption, XML parsing, and JSON mapping.


TEKsystems
Global Services
WORKFORCE
DEVELOPMENT


TEKsystems
Global Services
WORKFORCE
DEVELOPMENT

MODULE 35 OVERVIEW

Connect the React frontend to the Spring Boot CRM API -- typed HTTP clients, request states, error handling, and secure integration.

Theory: 1 Hour · Lab: 2 Hours · Total: 3 Hours -- confirmed against the official lesson-plan spreadsheet (Module 35 row).

DURATION & LAB

1 Hour Theory

Client-server communication, the request-response lifecycle, Axios, API responses, error handling, and loading states.

2 Hours Lab

Lab 35: wire the React CRM SPA to a real Spring Boot customer API with a typed fetch client.

Scenario

Freeze the SPA <-> API contract: typed fetch, AbortController cleanup, explicit request states, CORS, and failure-class tests.

MODULE ARC

Master Axios & Fetch

Send GET, POST, PUT, and DELETE requests and handle their responses correctly.

Handle Failure Gracefully

Loading, empty, error, and data states -- plus network, 4xx, and 5xx failure classes.

Build the Lab

Connect lab35-crm to a running Spring Boot API with CORS, correlation IDs, and tests.

KEY TAKEAWAY Total time: 3 Hours (1 Hour Theory + 2 Hours hands-on Lab). By the end, you will connect a React frontend to a Spring Boot REST API with production-grade discipline.

WHY API INTEGRATION MATTERS

APIs connect your frontend applications to powerful backend services and data.

WHY API INTEGRATION MATTERS

Enables Separation of Concerns — keeps frontend and backend independent, improving maintainability and scalability.

Improves Scalability — APIs allow services to scale independently based on demand.

Enhances Reusability — APIs expose functionality that can be reused across multiple clients and platforms.

Supports Multiple Clients — the same API can power web apps, mobile apps, and third-party integrations.

Faster Development — frontend teams can work in parallel with backend teams using well-defined APIs.

Strengthens Security — centralized business logic and validation on the server reduce client-side vulnerabilities.

THE ROLE OF APIS IN MODERN APPLICATIONS

KEY TAKEAWAY

API integration is the bridge between your user interface and your business logic. It enables communication, data exchange, and delivers seamless experiences to your users.

KEY TAKEAWAY API integration is the bridge between your user interface and your business logic -- it enables communication, data exchange, and seamless user experiences.

Why API Integration Matters

API

APIs connect different systems, services, and data, enabling **seamless communication** and unlocking **powerful capabilities.**

Connect
Integrate systems and applications easily

Innovate
Build new features faster with existing services

Save Time
Reduce development time and avoid rebuilding

Improve Experience
Deliver consistent and seamless user experiences

Key Benefits

Interoperability
APIs allow different platforms and technologies to work together without compatibility issues.

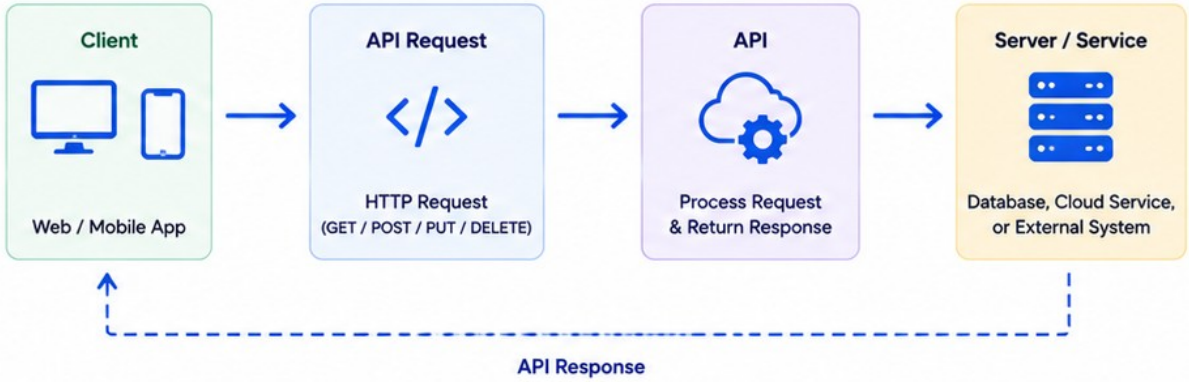
Faster Development
Leverage existing APIs and services to build and deploy applications more quickly.

Access to Data
Easily access real-time data and functionality from external systems and third-party services.

Cost Efficiency
Reduce costs by reusing existing services and minimizing duplicate efforts.

Scalability
APIs enable scalable solutions by integrating cloud services and microservices.

How API Integration Works



Bottom Line:
API integration drives efficiency, innovation, and growth by connecting systems, sharing data, and enabling seamless digital experiences.

Enable Innovation | Drive Business Growth | Deliver Greater Value

MODULE LEARNING OBJECTIVES

By the end of this module, you will be able to connect a modern React application to a robust Spring Boot API and build dynamic, data-driven user experiences.

1. **Understand the Lifecycle —** explain client-server communication and the request-response lifecycle.

2. **Use Axios —** send GET, POST, PUT, and DELETE requests to RESTful APIs.

3. **Handle Responses —** handle API responses and manage data effectively in frontend applications.

4. **Implement Error Handling —** implement error handling and manage loading states to improve user experience.

5. **Integrate with Spring Boot —** integrate React applications with Spring Boot REST APIs using best practices.

6. **Apply Security Best Practices —** apply security best practices when communicating between frontend and backend.

MODULE FOCUS AREAS

Axios & Fetch | **Response & Error Handling** | **Loading States** | **Spring Boot Integration** | **Frontend Security**

KEY TAKEAWAY You will build practical skills to connect modern React applications with robust Spring Boot APIs, creating dynamic, data-driven user experiences.

FRONTEND-BACKEND ARCHITECTURE

A frontend-backend architecture separates the user interface from the server-side logic and data management.

HOW IT WORKS

1. **User** — Interacts with the UI
2. **Frontend** — Sends HTTP request
3. **Backend** — Processes the request
4. **Data** — Fetched or updated
5. **Backend** — Returns JSON response
6. **Frontend** — Updates the UI

KEY BENEFITS

Separation of Concerns — UI/UX and business logic are developed and maintained independently.

Scalability — frontend and backend can scale independently based on demand.

Flexibility — multiple frontends (web, mobile, desktop) can consume the same backend APIs.

Maintainability — easier to update, test, and deploy components independently.

Better Team Collaboration — frontend and backend teams work in parallel with clear contracts (APIs).

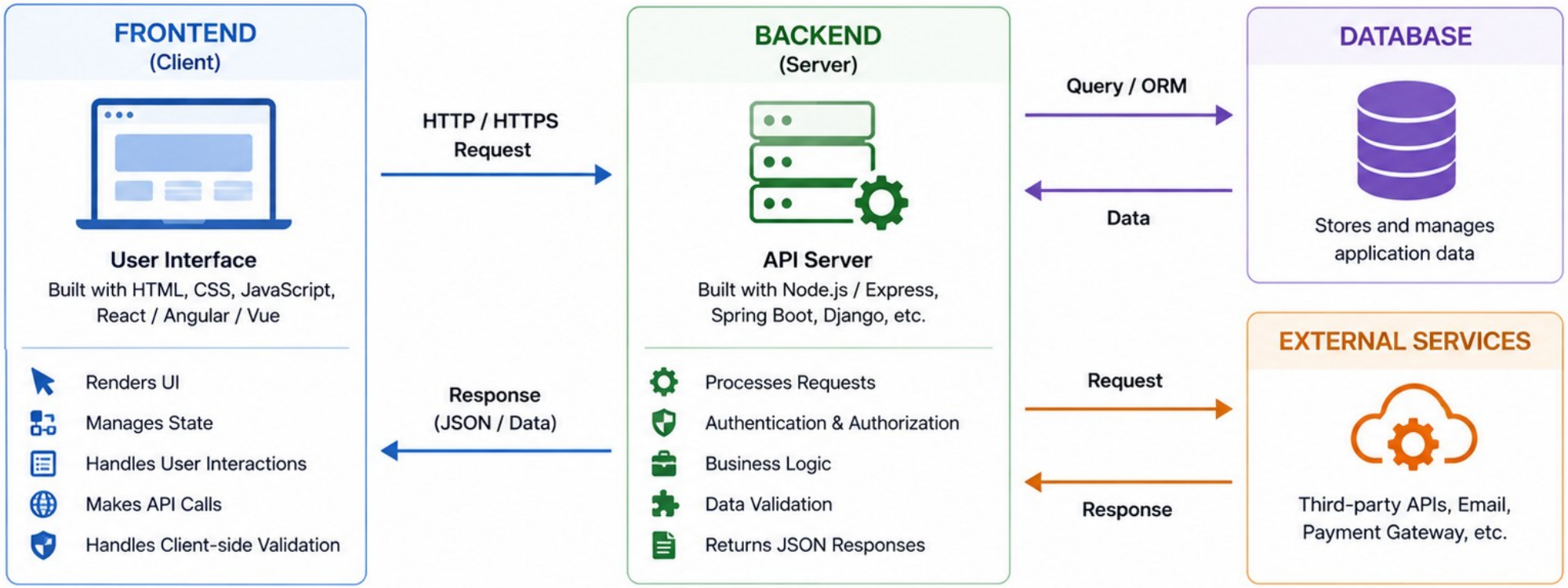
ARCHITECTURE LAYERS

Layer	Responsibilities
Frontend	Components, State Management, Routing, UI/UX (React)
Backend	Controllers, Services, Repositories, Security, Validation (Spring Boot)
Data Layer	Database (PostgreSQL / MySQL), other services (email, storage, third-party APIs)

KEY TAKEAWAY This architecture promotes modularity, reusability, and scalability while delivering a seamless experience to users.

Frontend-Backend Architecture

Separation of concerns between client-side (frontend) and server-side (backend)



KEY BENEFITS

Separation of Concerns
Frontend handles UI, backend handles logic & data.

Scalability
Frontend and backend can scale independently.

Flexibility
Use any frontend with any backend.

Maintainability
Easier to update and manage each layer.

Reusability
Backend APIs can be used by multiple clients.

REQUEST-RESPONSE LIFECYCLE

Every interaction between the frontend and backend follows a structured request-response lifecycle.

REQUEST FLOW

1. **1. User Action**
   clicks a button
2. **2. Frontend Sends**
   HTTP request
3. **3. Backend Receives**
   and routes it
4. **4. Process Request**
   business logic + data

RESPONSE FLOW

5. **5. Generate Response**
   usually JSON
6. **6. Send Response**
   back to frontend
7. **7. Frontend Receives**
   and parses data
8. **8. Update UI**
   shown to the user

KEY POINTS

Stateless | HTTPS Only | JSON Format | Error Handling

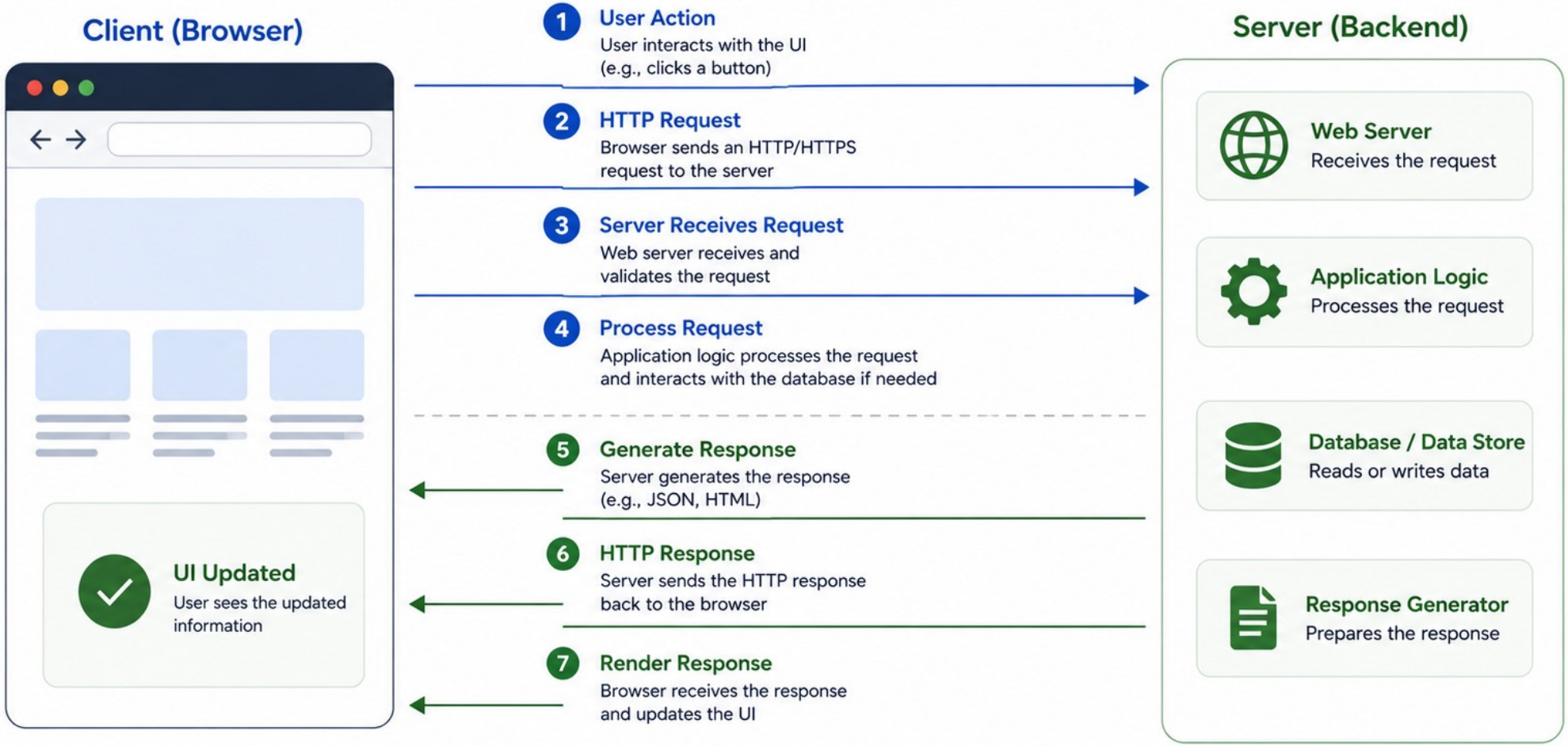
EXAMPLE -- "VIEW USERS"

- Click → GET /api/users → backend queries the database → JSON response → Axios receives it → React state updates → list is displayed.

KEY TAKEAWAY Understanding this lifecycle is key to building reliable, efficient, and user-friendly applications.

Request-Response Lifecycle

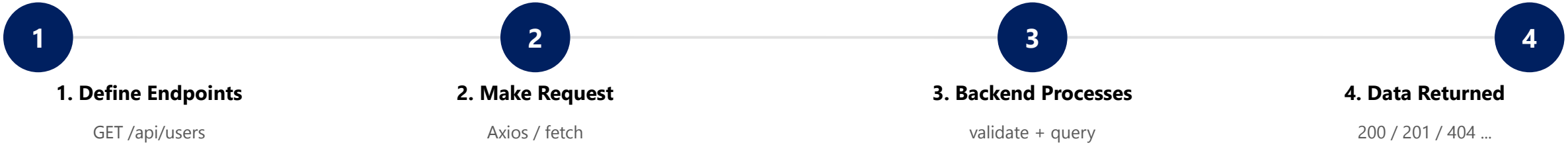
The journey of a client request and server response in a web application.



API INTEGRATION WORKFLOW (1 OF 2)

A structured workflow ensures seamless communication between the frontend application and backend APIs -- steps 1-4: from defining the endpoint to getting data back.

STEPS 1-4: FROM ENDPOINT TO RESPONSE



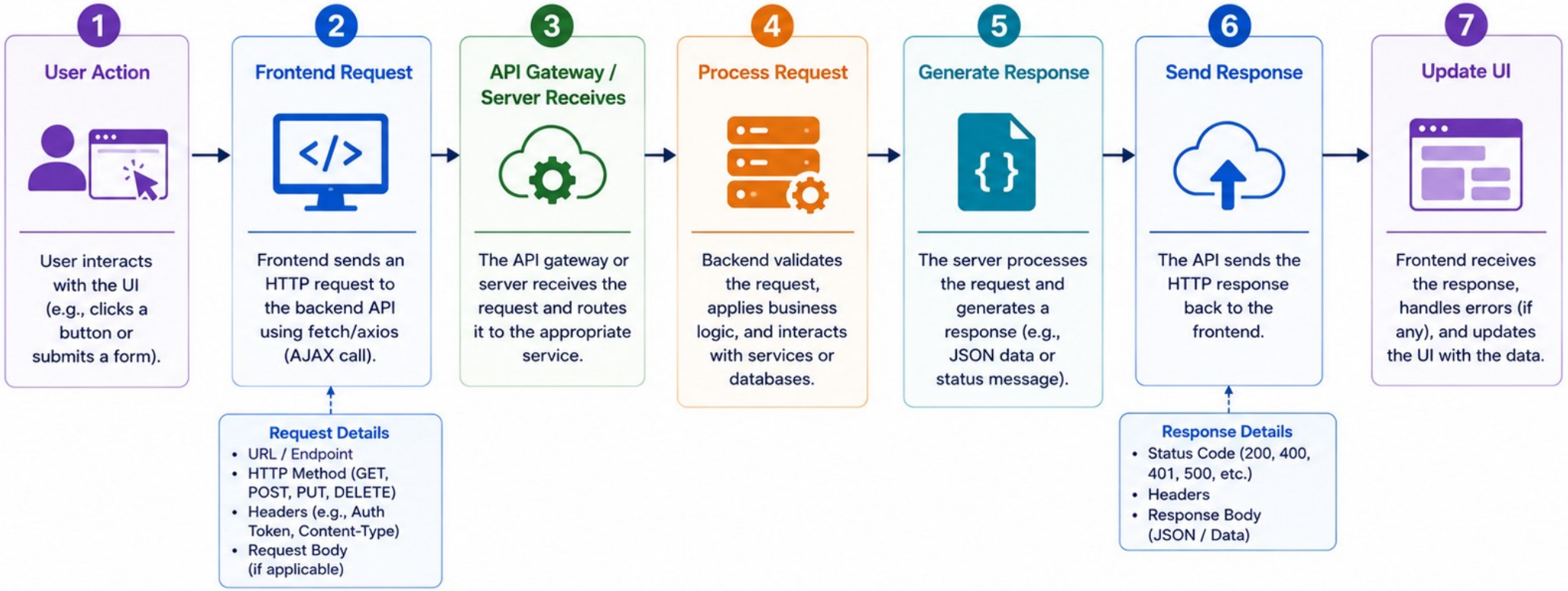
REAL-WORLD EXAMPLE

- User clicks “View Users” → React sends GET /api/users.
- Spring Boot fetches users from the database and returns JSON.
- Axios receives the response; data is set into React state.
- The user list is displayed on the screen.

KEY TAKEAWAY The first half of the workflow is about getting a well-defined request out the door and a response back -- define the endpoint, make the request, let the backend process it, and get data back with a meaningful status code.

API Integration Workflow

Connecting a frontend application with a backend API



Key Benefits

Separation of Concerns
Frontend focuses on UI/UX, backend handles logic and data.

Scalability
Frontend and backend can scale independently.

Security
Secure communication with authentication and authorization.

Reusability
APIs can be reused by multiple clients and platforms.

Maintainability
Easier updates and maintenance on both frontend and backend.

API INTEGRATION WORKFLOW (2 OF 2)

Steps 5-7: from receiving the response to a seamless UI update -- plus the best practices that keep the whole workflow reliable.

STEPS 5-7: FROM RESPONSE TO UPDATED UI

1

5. Frontend Receives

response data

2

6. Handle Data & Errors

try / catch

3

7. Update UI

seamless UX

BEST PRACTICES

- **Meaningful endpoint names**
- **Use HTTPS**
- **Handle loading states**
- **Handle errors gracefully**

KEY TAKEAWAY

A well-defined API integration workflow ensures efficient communication, better error handling, and a smooth user experience.

KEY TAKEAWAY Keep API contracts consistent and well-documented -- from defining endpoints to updating the UI, every step should be predictable and repeatable.

INTRODUCTION TO AXIOS

Axios is a promise-based HTTP client for the browser and Node.js. It makes it easy to send asynchronous HTTP requests to RESTful APIs and handle responses.

KEY FEATURES OF AXIOS

Promise Based — built on top of Promises, making async code easier to write and manage.

Easy to Use — simple and intuitive API for making HTTP requests.

Interceptors — intercept requests or responses before they are handled.

Error Handling — automatically handles errors and provides meaningful messages.

Browser & Node.js Support — works seamlessly in both browsers and Node.js environments.

INSTALLATION

```
npm install axios
```

BASIC EXAMPLE

```
import axios from 'axios';

// Send a GET request
axios.get('https://api.example.com/users')
  .then(response => {
    console.log('Data:', response.data);
  })
  .catch(error => {
    console.error('Error:', error);
  });
```

WHY USE AXIOS?

- Simplifies requests
- Handles JSON automatically
- Interceptors
- Better error handling

KEY TAKEAWAY Axios provides a clean, simple, and powerful way to interact with APIs, making it an essential tool for frontend development.

Introduction to Axios

Axios is a Promise-based HTTP client for JavaScript. It helps you make HTTP requests from the browser and Node.js with ease.

SENDING GET REQUESTS

GET requests are used to retrieve data from the server. They should not modify any data on the server.

SYNTAX

```
axios.get(url[, config])
```

EXAMPLE -- GET WITH QUERY PARAMETERS

```
axios.get('https://api.example.com/users', {
  params: { page: 1, size: 10 }
})
  .then(response => console.log(response.data))
  .catch(error => console.error('Error:', error));
```

COMMON STATUS CODES

200 OK **400 Bad Request** **401 Unauthorized**

404 Not Found **500 Server Error**

RESPONSE STRUCTURE

Property	Meaning
data	The response data returned from the server
status	HTTP status code (e.g., 200)
statusText	Status message (e.g., OK)
headers	Response headers
config / request	Original request config / request instance

BEST PRACTICES

- Use meaningful, RESTful endpoint names.
- Use query parameters for filtering, sorting, and pagination.
- Avoid exposing sensitive data in URLs.
- Show loading indicators while fetching data.

KEY TAKEAWAY Use axios.get() to retrieve data from the server. Handle the response and errors gracefully to build reliable applications.

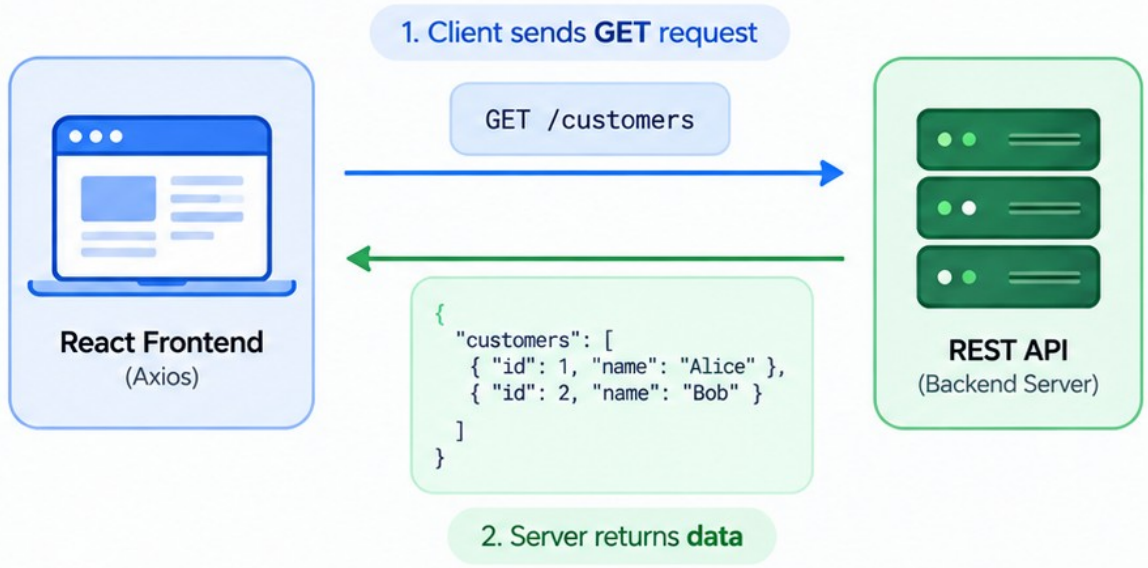
Sending GET Requests

Use Axios to fetch data from a server with an HTTP GET request.

React + Axios

```
import axios from 'axios';

const fetchCustomers = async () => {
  try {
    const response = await axios.get(
      'https://api.example.com/customers'
    );
    console.log(response.data); // Server response
  } catch (error) {
    console.error('Failed to fetch data:', error);
  }
};
```



Key Benefits

- Simple syntax
- Automatic JSON parsing
- Works in Browser and Node.js
- Promise-based (easy to use with async/await)

SENDING POST REQUESTS

POST requests are used to send data to the server to create new resources.

SYNTAX

```
axios.post(url, data, config)
```

EXAMPLE -- POST WITH HEADERS

```
const customer = {
  fullName: 'Amina Khan',
  email: 'amina@example.com',
  status: 'ACTIVE',
};
axios.post('https://api.example.com/customers', customer, {
  headers: { 'Content-Type': 'application/json' },
})
  .then(response => console.log(response.data))
  .catch(error => console.error('Error:', error));
```

COMMON STATUS CODES

201 Created **400 Bad Request** **409 Conflict** **500 Server Error**

EXAMPLE RESPONSE (201 CREATED)

```
{
  "data": {
    "id": "CUS-1001",
    "fullName": "Amina Khan",
    "email": "amina@example.com",
    "status": "ACTIVE"
  },
  "status": 201,
  "statusText": "Created"
}
```

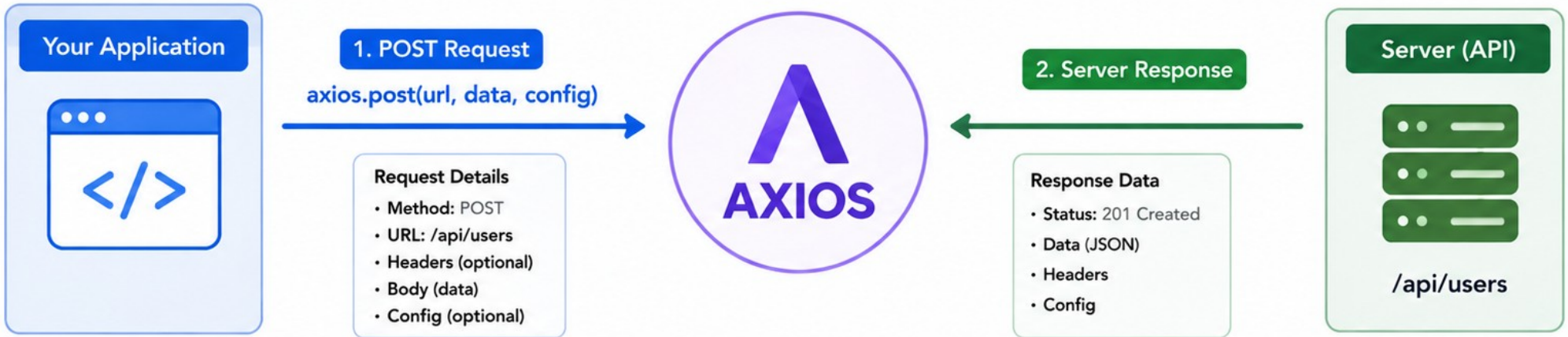
BEST PRACTICES

- Validate data on both frontend and backend.
- Always send data in JSON format.
- Use HTTPS and authentication for secure communication.
- Disable the submit control while the request is in flight.

KEY TAKEAWAY Use axios.post() to send data to the server and create new resources. Always handle responses and errors gracefully -- and guard against duplicate submits.

Sending POST Requests

Use **axios.post()** to send data to an API and create or update resources.



Example: Create User

```
import axios from 'axios';
const user = {
   name: 'John Doe',
   email: 'john@example.com',
   role: 'user'
};
axios.post('https://api.example.com/users', user, {
   headers: { 'Content-Type': 'application/json' }
})
 .then(response => {
    console.log('User created:', response.data);
    console.log('Status:', response.status);
})
 .catch(error => {
    console.error('Error:', error.response?.data || error.message);
});
```

Example Response

```
{
  "id": 101,
  "name": "John Doe",
  "email": "john@example.com",
  "role": "user",
  "createdAt": "2024-05-25T10:30:00Z"
}
```

Status: 201 Created

```
Headers:
Content-Type: application/json
```

SENDING PUT REQUESTS

PUT requests are used to update an existing resource on the server. They replace the entire resource with the new data.

SYNTAX

```
axios.put(url, data, config)
```

EXAMPLE -- UPDATE A CUSTOMER

```
const draft = {
  fullName: 'Ravi Singh',
  email: 'ravi@example.com',
  status: 'ACTIVE',
};
axios.put(
  `https://api.example.com/customers/${id}`,
  draft
)
  .then(response => console.log(response.data));
```

COMMON STATUS CODES

200 OK | 204 No Content | 400 Bad Request | 404 Not Found

RESPONSE STRUCTURE

Property	Meaning
data	The updated resource returned from the server
status	200 (OK) or 204 (No Content)

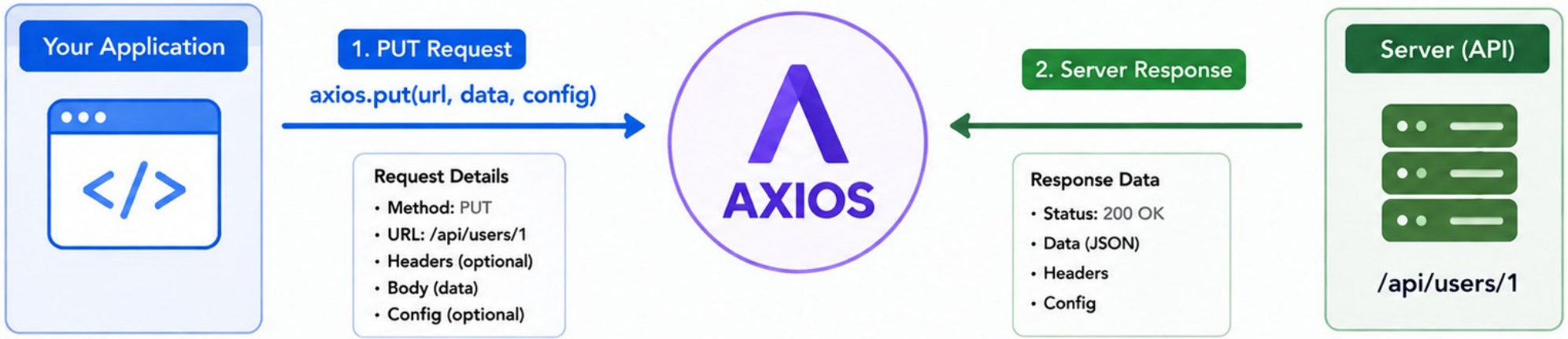
BEST PRACTICES

- Use PUT when you want to replace the entire resource.
- Ensure the resource exists before updating (encodeURIComponent the id).
- Validate data on both frontend and backend.
- Prefer the server-returned record over an optimistic-only update.

KEY TAKEAWAY Use axios.put() to update existing resources on the server. It replaces the entire resource with the new data.

Sending PUT Requests

Use **axios.put()** to update an existing resource on the server.



Example: Update User

```
import axios from 'axios';
const updatedUser = {
   name: 'John Doe',
   email: 'john.doe@example.com',
   role: 'admin'
};
axios.put('https://api.example.com/users/1', updatedUser, {
   headers: { 'Content-Type': 'application/json' }
})
 .then(response => {
    console.log('User updated:', response.data);
    console.log('Status:', response.status);
})
 .catch(error => {
    console.error('Error:', error.response?.data || error.message);
});
```

Example Response

```
{
  "id": 1,
  "name": "John Doe",
  "email": "john.doe@xample.com",
  "role": "admin",
  "updatedAt": "2024-05-25T12:45:00Z"
}
```

Status: 200 OK

```
Headers:
Content-Type: application/json
```

SENDING DELETE REQUESTS

DELETE requests are used to remove a resource from the server. Once deleted, the resource is no longer available.

SYNTAX

```
axios.delete(url, config)
```

EXAMPLE -- DELETE A RESOURCE

```
axios.delete('https://api.example.com/customers/CUS-1002')
  .then(() => {
    console.log('Customer deleted successfully');
  })
  .catch(error => console.error('Error:', error));
```

BEST PRACTICES

- Confirm the resource exists before attempting to delete it.
- Use appropriate authentication and authorization.
- Use HTTPS to protect sensitive operations.
- Show a confirmation prompt in the UI before deleting.
- Log delete operations for auditing purposes.

RESPONSE STRUCTURE

Response	Meaning
200 OK	Deleted; response body may contain a confirmation message
204 No Content	Deleted; no body returned

COMMON STATUS CODES

200 OK **204 No Content** **401 Unauthorized**

404 Not Found

KEY TAKEAWAY Use axios.delete() to remove resources from the server. Always handle responses and errors carefully and confirm actions to prevent accidental data loss.

Sending DELETE Requests

Use **axios.delete()** to remove a resource from the server.

Example: Delete User

```
import axios from 'axios';

axios.delete('https://api.example.com/users/1', {
   headers: {
      'Authorization': 'Bearer token'
   }
})
 .then(response => {
    console.log('User deleted successfully');
    console.log('Status:', response.status);
})
 .catch(error => {
    console.error('Error:', error.response?.data || error.message);
});
```

Example Response

```
{
  "status": 200,
  "statusText": "OK",
  "headers": {
    "content-type": "application/json",
    "date": "Sat, 25 May 2024 10:30:00 GMT"
  },
  "data": null,
  "config": { ... }
}
```

Status: 200 OK

DELETE requests often return 200 OK or 204 No Content when the resource is successfully removed.

HANDLING API RESPONSES

After sending a request, the API returns a response. Properly handling the response is crucial for a great user experience and robust applications.

RESPONSE FLOW

1. Sent
2. Processing
3. Returned
4. Handled

4xx Client Error (404)

```
{
  "status": 404,
  "statusText": "Not Found",
  "error": "RESOURCE_NOT_FOUND",
  "message": "User not found"
}
```

5xx Server Error (500)

```
{
  "status": 500,
  "statusText": "Internal Server
Error",
  "error": "INTERNAL_SERVER_ERROR",
  "message": "Something went wrong"
}
```

COMMON HTTP STATUS CODES

- 200 OK
- 201 Created
- 400 Bad Request
- 404 Not Found
- 500 Server Error

HANDLE RESPONSES IN REACT (EXAMPLE)

```
const fetchUsers = async () => {
  try {
    const response = await axios.get('/api/users');
    console.log('Data:', response.data);
    setUsers(response.data);
  } catch (error) {
    if (error.response) {
      // Server responded with a status outside 2xx
      console.error('Status:', error.response.status);
      setError(error.response.data.message);
    } else if (error.request) {
      // Request made, no response received
      console.error('No response:', error.request);
      setError('Cannot reach the server.');
    } else {
      // Something else happened
      console.error('Error:', error.message);
      setError('An unexpected error occurred.');
    }
  }
};
```

KEY TAKEAWAY Always handle API responses by checking status codes, processing data, and managing errors effectively to build reliable user experiences.

Handling API Responses

Process successful responses and handle errors gracefully using Axios.

Success Path (2xx)

```
axios.get('/api/data')
  .then(response => {
    // Handle success
    console.log(response.data);
  })
  .catch(error => {
    // Handle error
  });
```

Error Path (4xx, 5xx)

```
axios.get('/api/data')
  .then(response => {
    // Success
  })
  .catch(error => {
    // Handle error
    if (error.response) {
      console.error('Server responded with error:',
                    error.response.status);
      console.error('Message:', error.response.data);
    } else if (error.request) {
      console.error('No response received:';
                    error.request);
    } else {
      console.error('Request setup error:',
                    error.message);
    }
  });
```

Common Status Codes

200	OK	Request successful
201	Created	Resource created
204	No Content	Request successful, no content
400	Bad Request	Invalid request
401	Unauthorized	Authentication required
403	Forbidden	Access denied
404	Not Found	Resource not found
500	Internal Server Error	Server error
503	Service Unavailable	Service temporarily unavailable

Axios Response Object

```
response = {
  data: {...},
  status: 200,
  statusText: 'OK',
  headers: {...},
  config: {...},
  request: {...}
}
```

data	The response payload from the server.
status	The HTTP status code (e.g., 200, 404).
statusText	The status message (e.g., 'OK', 'Not Found').
headers	The response headers sent by the server.
config	The configuration used for the request.
request	The request object. (XHR in browser)

Axios Error Object

```
error = {
  message: 'Request failed with status code 404',
  name: 'AxiosError',
  stack: '...',
  config: {...},
  code: 'ERR_BAD_REQUEST',
  request: {...},
  response: {
    data: {...},
    status: 404,
    statusText: 'Not Found',
    headers: {...},
    config: {...}
  }
}
```

message	Error message description.
response	The server response (if available).
request	The request made (if no response).

Best Practices

- Always handle errors using .catch().
- Check error.response for server errors.
- Provide user-friendly messages.
- Log errors for debugging.
- Use interceptors for global error handling.

ERROR HANDLING IN FRONTEND APPLICATIONS

Errors can happen anywhere: network issues, server errors, invalid responses, or unexpected bugs. Good error handling improves user experience and reliability.

TYPES OF ERRORS

Network Errors — no internet, request timeout, DNS failure.

Server Errors — API returns 4xx (client) or 5xx (server) status codes.

Client/Validation Errors — invalid input, missing fields, business-rule violations.

Unexpected Errors — bugs in code, undefined states, parsing issues.

HANDLING API ERRORS (BEST PRACTICES)

- Always handle errors in every API call.
- Provide a retry option for recoverable errors.
- Show loading state and disable actions while loading.
- Log errors for debugging (e.g., Sentry, LogRocket).
- Never expose sensitive information in error messages.

SHOWING USER-FRIENDLY MESSAGES

Be Clear — use simple language and avoid technical jargon.

Be Helpful — suggest what the user can do next (e.g., Retry).

Be Contextual — show messages near the relevant field or section.

Be Dismissible — allow the user to close or retry the message.

EXAMPLE UI

Message shown to the user
⚠ **Failed to load customers. Please try again.**
⚠ **Please check your input and try again.**
⚠ **No internet connection. Check your network.**

KEY TAKEAWAY Handling errors gracefully makes your application trustworthy, user-friendly, and production-ready.

Error Handling in Frontend Applications

Detect, handle, and communicate errors gracefully to improve user experience.

Error Handling Flow

1. Error Occurs
An error happens during API call, rendering, or user interaction.

2. Detect Error
Catch the error using try…catch, promises, or error boundaries.

3. Handle Error
Log the error, show friendly message, and decide recovery action.

4. Inform User
Display user-friendly feedback (toast, alert, or UI message).

5. Recover / Retry
Allow retry, fallback UI, or redirect based on the error type.

1. Try–Catch with Async/Await

```
const fetchData = async () => {
  try {
    const response = await axios.get('/api/data');
    setData(response.data);
  } catch (error) {
    console.error('Failed to fetch data:', error);
    setError('Unable to load data. Please try again.');
  }
};
```

Best Practices

- Always show user-friendly messages.
- Log technical details for debugging.
- Use specific messages for different error types.
- Provide retry or recovery options.
- Keep the UI consistent even on errors.

2. Error Boundaries for UI Errors

```
class ErrorBoundary extends React.Component {
  constructor(props) {
    super(props);
    this.state = { hasError: false };
  }
  static getDerivedStateFromError() {
    return { hasError: true };
  }
  componentDidCatch(error, info) {
    console.error('UI Error:', error, info);
  }
  render() {
    if (this.state.hasError) {
      return <FallbackUI message="Something went wrong." />;
    }
    return this.props.children;
  }
}
```

Use for unexpected UI errors
(rendering, lifecycle, component errors).

3. Handle Different Error Types

Network Errors — No internet, timeout, DNS issues.	Show: "Check your connection."
Server Errors (5xx) — Issues on the server side.	Show: "Server error. Please try again later."
Client Errors (4xx) — Invalid request, unauthorized, not found.	Show: "Invalid request or access denied."
Validation Errors — Invalid input from user.	Show: Field-specific validation messages.
Unknown Errors — Unexpected or undefined errors.	Show: "Something went wrong."

User Experience Tips

- Be clear and friendly
- Offer retry options
- Keep users informed
- Ensure accessibility of messages
- Don't expose sensitive technical details

TRY IT YOURSELF — EXERCISE 1: ERROR UX COPY

Reinforces: user-facing failure messages for 404, network, and 400 CRM API errors -- a documentation exercise.

DRAFT USER-FACING MESSAGES (RECORD IN notes/error-ux.md)

Step	What To Do
1 -- 404 Message	Draft the message shown when GET /api/customers/CUS-9999 returns 404: the customer was not found.
2 -- Network Message	Draft the message shown when the API is unreachable -- no response, a timeout, or a CORS rejection.
3 -- 400 Message	Draft the message shown when a create/update request fails name validation with a 400 Bad Request.
4 -- Logging vs UX Boundary	Write the rule: the dev console may log the X-Correlation-Id and status code -- the user only ever sees plain language.

PASS CRITERIA

404 / network / 400 messages -- all three are drafted in plain language, each naming what happened without jargon.

Correlation stays in logs -- X-Correlation-Id and raw status codes are noted as dev-console/log detail, never shown to the user.

No stack traces in UI copy -- none of the three messages leak a stack trace, SQL error, or internal exception name.

KEY TAKEAWAY TRY IT NOW -- Complete Exercise 1 before continuing: exercises/exercise-01-error-ux.md (workspace: examples/module-35-exercises).

LOADING STATES AND USER EXPERIENCE

Loading states inform users that something is happening. They reduce uncertainty, prevent duplicate actions, and create a smooth, professional experience.

WHY LOADING STATES MATTER

Reduce Uncertainty — users know the app is working and their action was received.

Prevent Duplicate Actions — disable buttons or show loaders to avoid multiple submissions.

Improve Perceived Performance — users are more patient when kept informed.

Enhance Professionalism — good UX builds trust and confidence in your application.

IMPLEMENTING LOADING STATES (REACT + FETCH)

```
const [customers, setCustomers] = useState([]);
const [loading, setLoading] = useState(false);
const [error, setError] = useState(null);

const fetchCustomers = async () => {
  setLoading(true);
  setError(null);
  try {
    const data = await customersApi.list();
    setCustomers(data);
  } catch (err) {
    setError(err.message);
  } finally {
    setLoading(false); // always hide the loader
  }
};
```

EXAMPLE: USER LIST COMPONENT

Loading	Success	Error
Skeleton rows / spinner	Customer cards rendered	⚠ Failed to load. Retry

COMMON LOADING UI PATTERNS

Spinner **Skeleton Screen** **Button Loader** **Progress Bar**

BEST PRACTICES

Show loading immediately **Disable inputs while loading** **Never hide errors**

KEY TAKEAWAY Well-designed loading states keep users informed, reduce frustration, and create a smoother, more reliable experience.

Loading States and User Experience

Good loading experiences keep users informed, reduce uncertainty, and build trust.

Why Loading States Matter

Set Expectations
Let users know something is happening.

Reduce Frustration
Prevent users from thinking the app is broken.

Improve Perceived Performance
Feedback makes wait time feel shorter.

Build Trust
A responsive UI feels reliable and polished.

Common Loading Indicators

Spinner / Activity Indicator
Good for short waits.

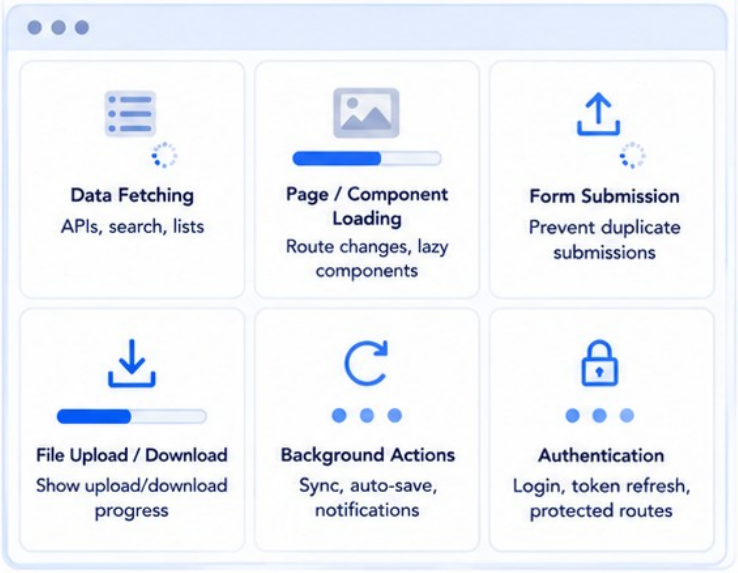
Progress Bar
60%
Show progress for longer tasks or known durations.

Dots / Typing Indicator
Subtle indication of activity.

Skeleton Screen
Show placeholder content that mimics the final layout.

Shimmer Effect
Animate placeholders to indicate content is loading.

Where to Use Loading States



Data Fetching
APIs, search, lists

Page / Component Loading
Route changes, lazy components

Form Submission
Prevent duplicate submissions

File Upload / Download
Show upload/download progress

Background Actions
Sync, auto-save, notifications

Authentication
Login, token refresh, protected routes

Best Practices

Keep it Immediate
Show loading states instantly when an action starts.

Be Honest
Avoid fake progress. Only show progress when real.

Keep it Contextual
Place loaders close to the content being loaded.

Avoid Blocking the Whole UI
Use inline loaders or skeletons instead of full-page blockers when possible.

Add Time Thresholds
Show loaders only if the task takes longer than ~200–300ms to avoid flicker.

Provide Feedback
Combine loaders with helpful text like "Loading...", "Please wait...", or estimated time.

Example: Good Loading Experience Flow

User initiates action
e.g., click, search, submit

→

Show appropriate loading indicator
Immediate feedback

→

Process in background
Fetch data, save, or perform task

→

Show results or success state
Update UI with data

→

Great experience
User stays informed and satisfied

TRY IT YOURSELF — EXERCISE 2: FETCH FLOW

Reinforces: the idle/loading/success/error state machine for listing customers, including abort and empty-state UX -- an architecture exercise.

SKETCH THE STATE MACHINE (RECORD IN notes/fetch-flow.md)

```
idle -> loading -> success | error

useEffect(() => {
  const controller = new AbortController();
  setState('loading');
  customersApi
    .list({ signal: controller.signal })
    .then((data) => setState('success', data))
    .catch((err) => setState('error', err));
  return () => controller.abort(); // cleanup
}, []);
```

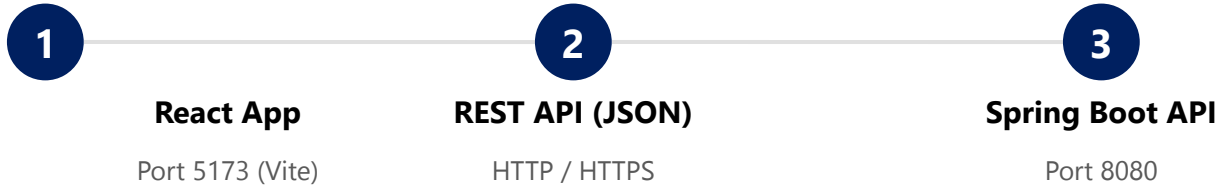
#	What To Do
1	Name the four states for the customer list view: idle, loading, success, error.
2	Sequence: mount -> set loading -> fetch -> set data (Amina + Ravi) or set an error message.
3	Note AbortController cleanup on unmount, so a late response can't setState after navigate-away.
4	Draft empty-state copy for when the API returns [] -- distinct from both loading and error.

KEY TAKEAWAY TRY IT NOW -- Complete Exercise 2 before continuing: exercises/exercise-02-fetch-flow.md (workspace: examples/module-35-exercises).

CONNECTING REACT TO SPRING BOOT (1 OF 2)

React (frontend) communicates with Spring Boot (backend) through REST APIs. This connection enables data exchange and builds full-stack applications.

APPLICATION ARCHITECTURE



CONNECTION STEPS

- Start the Spring Boot application (usually on port 8080).
- Expose REST endpoints that return JSON.
- Start the React (Vite) development server (usually on port 5173).
- Configure the API base URL via a Vite environment variable.
- Use fetch or Axios in React to call REST endpoints.
- Spring Boot processes the request and returns JSON.
- React receives the response and updates the UI.

CROSS-ORIGIN RESOURCE SHARING (CORS)

React and Spring Boot run on different origins (ports). CORS must be enabled on the backend so the browser allows the request.

WebConfig.java

```
@Configuration
public class WebConfig implements WebMvcConfigurer {
  @Override
  public void addCorsMappings(CorsRegistry registry) {
    registry.addMapping("/api/**")
        .allowedOrigins("http://localhost:5173")
        .allowedMethods("GET","POST","PUT","DELETE")
        .allowedHeaders("Content-Type","X-Correlation-Id");
  }
}
```

BEST PRACTICES

Env vars for base URL

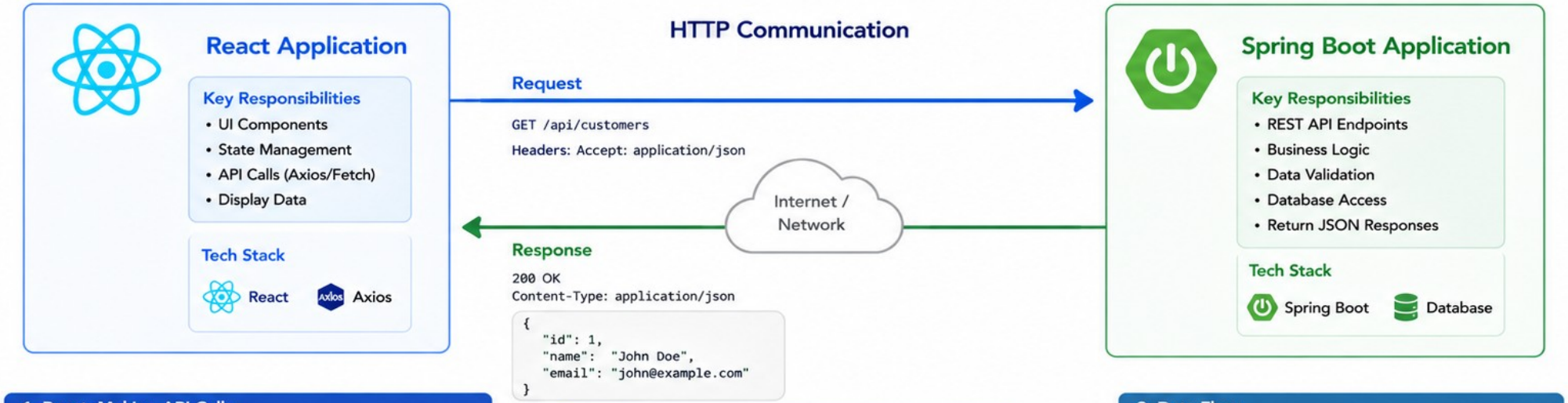
Handle loading & error states

Allowlist specific origins only

KEY TAKEAWAY React communicates with Spring Boot using REST APIs over JSON. A restricted CORS allowlist and an environment-configured base URL keep the connection correct and secure.

Connecting React to Spring Boot

React frontend communicates with Spring Boot backend via REST APIs (JSON over HTTP).



1. React: Making API Calls

Using Axios

```
import axios from 'axios';
const API_BASE = 'http://localhost:8080/api';

export const fetchCustomers = async () => {
  try {
    const response = await axios.get('${API_BASE}/customers');
    return response.data;
  } catch (error) {
    console.error('Error fetching customers:', error);
    throw error;
  }
};
```

2. Spring Boot: REST API Endpoint

Example Controller

```
@RestController
@RequestMaping("/api/customers")
public class CustomerController {
  private final CustomerService service;

  public CustomerController(CustomerService service) {
    this.service = service;
  }

  @GetMapping
  public List<Customer> getAllCustomers() {
    return service.getAllCustomers();
  }
}
```

3. Data Flow

1. User interacts with React UI (e.g., clicks "Load Customers")
2. React makes API request using Axios
3. Spring Boot processes request and fetches data
4. Response returned as JSON — React displays the data

Best Practices

- Use environment variables for API base URL
- Enable CORS in Spring Boot for frontend origin
- Handle errors and loading states in React
- Use DTOs for clean API responses
- Secure APIs with authentication (JWT)

TRY IT YOURSELF — EXERCISE 3: CORS AND HEADERS

Reinforces: local CORS origins and the X-Correlation-Id header for the CRM SPA -- a documentation exercise.

DOCUMENT CORS + HEADERS (RECORD IN notes/cors-and-headers.md)

Step	What To Do
1 -- Origins	Record the two local origins: UI at http://localhost:5173 (Vite), API at http://localhost:8080 (Spring Boot) -- adjust if your lab differs.
2 -- Explain CORS	Write one sentence: the browser blocks cross-origin fetch/XHR calls unless Spring's CORS config explicitly allows the UI's origin.
3 -- Plan the Header	Note the header every fetch call carries: X-Correlation-Id: lab-request-001.
4 -- Secrets Boundary	Write the rule: never place DB passwords or other secrets in frontend env vars -- only the public API base URL belongs in VITE_CRM_API_URL.

PREFLIGHT SYMPTOM (BROWSER CONSOLE) -- WHAT A MISSING ALLOWLIST LOOKS LIKE

```
Access to fetch at 'http://localhost:8080/api/customers' from origin
'http://localhost:5173' has been blocked by CORS policy: No
'Access-Control-Allow-Origin' header is present on the requested resource.
```

KEY TAKEAWAY TRY IT NOW -- Complete Exercise 3 before continuing: exercises/exercise-03-cors-and-headers.md (workspace: examples/module-35-exercises).

CONNECTING REACT TO SPRING BOOT (2 OF 2)

A Spring Boot REST controller on one side, a typed fetch client on the other -- the same request/response contract, seen from both ends.

EXAMPLE: SPRING BOOT ENDPOINT

CustomerController.java

```java
@RestController
@RequestMapping("/api/customers")
public class CustomerController {

  @GetMapping
  public List<Customer> getAll() {
    return customerService.getAll();
  }

  @PostMapping
  public ResponseEntity<Customer> create(
      @Valid @RequestBody CustomerDraft draft) {
    Customer created = customerService.create(draft);
    return ResponseEntity.status(201).body(created);
  }
}
```

EXAMPLE: REACT FETCH CLIENT

api/customers.ts

```ts
const API_URL = import.meta.env.VITE_CRM_API_URL;

export async function request(path, init = {}) {
  const res = await fetch(`${API_URL}${path}`, {
    ...init,
    headers: {
      'Content-Type': 'application/json',
      'X-Correlation-Id': 'lab-request-001',
      ...init.headers,
    },
  });
  if (!res.ok) throw await ApiError.from(res);
  if (res.status === 204) return undefined;
  return res.json();
}
```

COMMON HTTP METHODS

GET → Retrieve | **POST → Create** | **PUT → Update** | **DELETE → Delete**

KEY TAKEAWAY @RestController exposes JSON endpoints on the Spring side; one typed request() helper owns headers, error translation, and 204 handling on the React side.

TRY IT YOURSELF — EXERCISE 4: ENDPOINT MAP

Reinforces: mapping CRM dashboard actions to Spring REST endpoints -- an analysis exercise.

UI ACTION -> ENDPOINT (RECORD IN notes/lab35-api.md)

UI Action	HTTP
List customers	GET /api/customers
Open Amina	GET /api/customers/CUS-1001
Open Ravi	GET /api/customers/CUS-1002
Create customer	POST /api/customers
Update status	PATCH /api/customers/{id}

STEPS

#	What To Do
1	Copy the reference table into notes/lab35-api.md; note any path differences from your Week 3 API.
2	Add the missing GET row for Ravi (CUS-1002).
3	List the five expected status codes: 200, 201, 400, 404, 500.
4	Sketch the list-item JSON shape: customerId, name, status.

LIST ITEM JSON SKETCH

```
{
  "customerId": "CUS-1001",
  "name": "Amina Khan",
  "status": "ACTIVE"
}
```

KEY TAKEAWAY TRY IT NOW -- Complete Exercise 4 before continuing: exercises/exercise-04-endpoint-map.md (workspace: examples/module-35-exercises).

TRY IT YOURSELF — EXERCISE 5: FILL FETCH TODOS (STARTER)

Reinforces: completing a typed fetch helper for customers -- fill in every blank before it will compile.

FILL IN EVERY ____ -- customers.ts (STARTER, notes/lab35-todos.md)

```
export type Customer = {
  customerId: string; name: string; status: string
};

export async function listCustomers(
  signal?: AbortSignal
): Promise<Customer[]> {
  const res = await fetch(_____, {
    headers: { "X-Correlation-Id": "_____" },
    signal,
  });
  if (!res.ok) throw new Error(`HTTP ${res.status}`);
  return (await res.json()) as _____;
}

export async function getCustomer(id: string) {
  const res = await fetch(`${_____}/${id}`);
  // TODO: handle 404 for unknown id
  return (await res.json()) as Customer;
}
```

#	What To Do
1	Fill the fetch URL and header blanks: "/api/customers" and "lab-request-001".
2	Fill the return-type blank with Customer[]; use the same base URL in getCustomer.
3	Add // TODO: on success setCustomers including Amina + Ravi fixtures from API.
4	Add // TODO: map 400 body.detail to form error string.

KEY TAKEAWAY TRY IT NOW -- This is a STARTER, not a solution: complete every blank in exercises/exercise-05-fill-fetch-todos.md, then verify it compiles/runs before continuing.

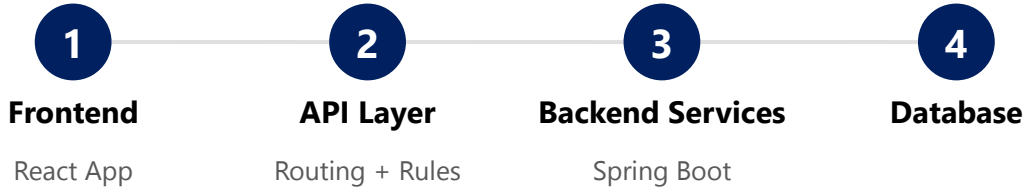
API LAYER DESIGN (1 OF 2)

A well-designed API layer acts as a bridge between frontend and backend, providing secure, reliable, and consistent access to business functionality.

WHAT IS AN API LAYER?

A dedicated layer that exposes application functionality through well-defined, secure, and standardized APIs -- the intermediary that handles requests, enforces rules, and returns properly structured responses.

ROLE OF THE API LAYER



API LAYER RESPONSIBILITIES

Routing — map incoming requests to the correct endpoint.

Validation — validate path variables, query params, headers, and body.

Authentication & Authorization — verify identity and permissions.

Business Logic Orchestration — coordinate with services to process requests.

Response Formatting — return consistent JSON responses and status codes.

Error Handling — handle exceptions and return meaningful error responses.

DESIGN PRINCIPLES

Consistency — consistent naming, data formats, and response structures across all endpoints.

Separation of Concerns — keep routing, validation, business logic, and data access separated.

Stateless & Resource-Oriented — each request is self-contained; APIs model business resources (nouns).

Security & Versioning — authenticate every request; version APIs for backward compatibility.

KEY TAKEAWAY A well-designed API layer ensures reliable communication, security, and scalability -- focus on consistency, security, and a great developer experience.

API Layer Design

Designing a clean, consistent, scalable, and secure API layer.

Design Principles

Resource-Oriented
Model APIs around business resources.

Consistent
Use consistent naming, structure, and status codes.

Stateless
Each request contains all the information needed.

Secure
Validate input, authenticate, and authorize every request.

Scalable
Design for evolution and backward compatibility.

REST Endpoints Example

Method	Endpoint	Description
GET	/api/customers	Get all customers
GET	/api/customers/{id}	Get customer by ID
POST	/api/customers	Create a new customer
PUT	/api/customers/{id}	Update customer by ID
DELETE	/api/customers/{id}	Delete customer by ID

API Layer Architecture

Controller Layer
- Handle HTTP requests
- Request validation
- Map requests to service layer
- Return appropriate responses

Service Layer
- Business logic
- Orchestrate operations
- Transaction management
- Call repository layer

Repository Layer
- Data access logic
- Interact with the database
- Use JPA / Spring Data

Database Layer
- Persistent data storage

Request / Response Structure

Request Example

```
{
  "name": "John Doe",
  "email": "john@example.com",
  "phone": "123-456-7890"
}
```

Response Example (Success)

```
{
  "status": "success",
  "message": "Customer created successfully",
  "data": {
    "id": 101,
    "name": "John Doe",
    "email": "john@example.com"
  }
}
```

Response Example (Error)

```
{
  "status": "error",
  "message": "Email already exists",
  "timestamp": "2024-05-01T10:30:00Z",
  "path": "/api/customers"
}
```

Best Practices

- Use meaningful and plural resource names.
- Follow HTTP status code standards.
- Implement pagination, filtering, and sorting.
- Version your APIs (e.g., /api/v1/...).
- Validate input and handle errors gracefully.
- Document APIs with OpenAPI/Swagger.
- Ensure authentication and authorization.

API Request Flow

Client sends HTTP Request → Controller receives request → Service processes business logic → Repository accesses data → Database returns data → Controller returns HTTP Response

API LAYER DESIGN (2 OF 2)

REST API design structure, a worked endpoint example, best practices, and common pitfalls to avoid.

REST API DESIGN STRUCTURE

Layer	Responsibility
Controller	Defines endpoints; handles HTTP requests/responses; validates input.
Service	Implements business logic; coordinates operations and transactions.
Repository	Data access via JPA; maps entities to persistent storage.

EXAMPLE ENDPOINT

GET /api/v1/customers/123

```
curl -X GET "http://localhost:8080/api/v1/customers/123" \
  -H "X-Correlation-Id: lab-request-001"

// 200 OK
{ "data": { "id": 123, "fullName": "Amina Khan" },
  "status": 200, "message": "Customer retrieved" }
```

BEST PRACTICES

- **Resource-based URIs, plural nouns**
- **Correct HTTP methods**
- **Meaningful status codes**
- **Use DTOs to control exposure**
- **Document with OpenAPI / Swagger**

COMMON PITFALLS

- Inconsistent URI naming conventions.
- Mixing business logic into controllers.
- Poor error handling and unclear responses.
- Exposing internal implementation details.
- Skipping API versioning.

KEY TAKEAWAY A well-designed API layer with clear contracts, consistent patterns, and robust error handling ensures a secure, scalable, and maintainable frontend-backend integration.

FRONTEND SECURITY CONSIDERATIONS

Security in frontend applications is critical to protect users, data, and your backend services. Remember: never trust the client -- always validate and enforce security on the server.

TOP SECURITY RISKS IN FRONTEND

Cross-Site Scripting (XSS) — attackers inject malicious scripts to steal data or perform actions.

Cross-Site Request Forgery (CSRF) — forces users to execute unwanted actions.

Sensitive Data Exposure — exposing tokens, keys, or PII in the client.

Insecure Storage — storing tokens or sensitive data in LocalStorage improperly.

Insecure API Calls — calling APIs without authentication or over HTTP.

CORS Misconfiguration — overly permissive CORS policies (e.g., allowedOrigins("*")).

SECURE FRONTEND PRACTICES

Sanitize User Input — encode or sanitize input before rendering.

Use HTTPS — always use HTTPS for API calls and static assets.

Store Tokens Securely — prefer HttpOnly cookies over localStorage for tokens.

Implement CSP — restrict sources for scripts, styles, images, and frames.

Validate & Authorize on the Server — never trust data from the client.

Handle Errors Safely — show user-friendly messages; never expose internal details.

SECURITY CHECKLIST

- **All API calls use HTTPS**
- **Tokens stored securely**
- **Input validated & sanitized**
- **Dependencies up to date**
- **CORS configured correctly**

KEY TAKEAWAY Frontend security is about reducing attack surface, protecting user data, and using secure practices at every step -- always combine frontend safeguards with strong backend security.

Frontend Security Considerations

Best practices to build secure and resilient frontend applications

1 Validate All Inputs

Validate and sanitize all user inputs on the client side to prevent malicious data injection.

2 Prevent XSS

Escape or sanitize dynamic content. Avoid rendering untrusted HTML content.

3 Secure API Calls

Always use HTTPS for API requests to protect data in transit.

4 Store Tokens Securely

Store tokens in httpOnly cookies or secure storage. Avoid localStorage for sensitive data.

5 Implement CSP

Use Content Security Policy to restrict sources of scripts, styles, images and more.

6 Prevent CSRF

Use anti-CSRF tokens for state-changing requests and validate on the server.

7 Secure Dependencies

Regularly update libraries and scan for vulnerabilities in frontend dependencies.

8 Limit Data Exposure

Do not expose sensitive information in the frontend or in API responses.

9 Use Secure Headers

Ensure the backend sets security headers to protect against common attacks.

10 Handle Errors Safely

Avoid exposing sensitive error details. Show generic messages to users.

Security is a shared responsibility. | Frontend security works hand-in-hand with backend security.

TRY IT YOURSELF — EXERCISE 6: LAB 35 READINESS CHECKLIST

Capstone pre-lab check -- confirm the API contract and scope boundaries before opening Lab 35.

READINESS CHECKLIST (RECORD IN notes/lab35-readiness.md)

Check	What To Confirm
Dependency	Lab 35 needs a running Spring API in the real lab -- confirm this pre-lab does NOT start Spring Boot or Vite.
Evidence Preview	Describe the Network-panel evidence you will capture in the lab: a GET list call plus the Amina (CUS-1001) detail call.
Security Boundary	Write one line: Authorization/bearer-token headers wait for Lab 36 -- do not implement JWT login UI yet.
Pass Mark	State the pre-lab pass mark: Exercise 4's fetch TODOs are filled and verified before Lab 35 begins.

SETUP (POWERSHELL) -- FROM EXERCISES-INDEX.MD

```
cd $env:USERPROFILE\java-bootcamp
New-Item -ItemType Directory -Force -Path examples\module-35-exercises | Out-Null
cd examples\module-35-exercises
java -version
```

KEY TAKEAWAY TRY IT NOW -- Complete Exercise 6 before opening Lab 35: exercises/exercise-06-lab35-readiness.md.

LAB OVERVIEW -- FRONTEND TO API INTEGRATION

Lab 35: Integrating React with the Spring CRM API -- lab35-crm

BUSINESS SCENARIO

- The CRM React client must talk to Spring Boot over HTTPS/JSON in production; Spring persists to PostgreSQL and may bridge SOAP internally, but the browser speaks JSON only.
- Leadership freezes the gate: no merge of SPA ↔ API integration without typed fetch, AbortController cleanup, explicit request states, a CORS allowlist, and tests for 2xx / 4xx / 5xx / network / abort.
- You own that gate for listing Amina (CUS-1001) and Ravi (CUS-1002), creating a draft, mapping email 400s, and rejecting https://evil.example.

LEARNING OBJECTIVES

- Document the REST (or SOAP-bridge) customer contract with curl evidence.
- Configure a public API base URL via Vite env (no secrets in VITE_*).
- Write a typed fetch client and a normalized ApiError.
- Render loading, success, empty, and error states distinctly.
- Cancel obsolete requests with AbortController.
- Map backend validation errors to labeled form fields; configure restricted CORS.

WHAT YOU BUILD

- lab35-crm/crm-ui: typed ApiError + http.request + customersApi; abortable list load with distinct loading/empty/data/error UI; create/update with X-Correlation-Id: lab-request-001 and a duplicate-submit guard; backend 400 email-field mapping; Spring CORS allowlist for localhost:5173; mock tests for 200/201/400/500/network/abort; evil-Origin curl evidence.

KEY TAKEAWAY Fixtures CUS-1001 (Amina Khan, ACTIVE) and CUS-1002 (Ravi Singh, PROSPECT) anchor every example; every mutating call carries X-Correlation-Id: lab-request-001; secrets never go in VITE_*.

LAB TASKS AND DELIVERABLES

lab35-crm -- implementation steps, deliverables, and the evaluation rubric.

#	Implementation Step
1	Inspect the API contract -- curl list/create; confirm CUS-1001/CUS-1002 JSON shapes.
2	Configure VITE_CRM_API_URL via .env.example (no secrets in VITE_*).
3	Create the normalized ApiError type (status, code, message, fieldErrors).
4	Build the http.request fetch boundary -- headers, 204 guard, non-OK → ApiError.
5	Create typed customersApi (list / create / update) on top of http.request.
6	Cancel obsolete loads with AbortController in the list-load effect.
7	Render distinct loading / empty / data / error states; map 400 field errors to the form.
8	Prevent duplicate POSTs (saving flag); restrict Spring CORS; test all response classes + probe evil Origin.

EXPECTED DELIVERABLES

- Typed ApiError / http / customersApi with abortable loads.
- Distinct loading / empty / data / error UX.
- Create/update with correlation header + duplicate-submit guard.
- Backend 400 field mapping.
- Spring CORS allowlist + evil-Origin evidence.
- Response-class tests + green build (twice).
- Integration notes + screenshots.
- README runbook (Spring + Vite); no secrets committed.

RUBRIC (100 MARKS)

- Environment 10 · Core Impl 30 · Integration/Config 15 · Failure Handling 15 · Verification 10 · Security/Prod Awareness 10 · Docs 10

KEY TAKEAWAY Swallowing errors into empty lists loses failure-handling marks; leaving allowedOrigins("*") in code is an honor violation -- prove abort, 400 mapping, and evil-Origin denial all work.

MODULE 35 SUMMARY

In this module, you connected a React frontend to a Spring Boot REST API, handled data and errors gracefully, and delivered a smooth user experience.

KEY CONCEPTS COVERED

Axios & the 4 HTTP Verbs

GET, POST, PUT, and DELETE requests to RESTful Spring Boot APIs, and their response shapes.

Request-Response Lifecycle

Client-server communication, the eight-step request/response flow, and status codes.

Responses & Error Handling

Parsing response objects and handling network, server, and validation failures gracefully.

Loading States

Distinct loading, empty, data, and error UX that keeps users informed and confident.

React + Spring Boot Integration

CORS, environment configuration, and a well-designed API layer connecting the two.

Frontend Security

Input sanitization, secure token storage, HTTPS, CSP, and server-side validation.

MODULE FLOW RECAP

1. **Axios / Fetch**
2. **4 HTTP Verbs**
3. **Handle Response**
4. **Loading & Error UX**
5. **CORS & Security**
6. **Lab: lab35-crm**

KEY TAKEAWAY Frontend-to-API integration is the seam where your UI meets your business logic -- typed clients, explicit states, and defensive error handling make that seam reliable.

KNOWLEDGE CHECK (1 OF 2)

Test your understanding of Axios, HTTP methods, and data formats used when connecting React to a Spring Boot REST API.

1. Which library is commonly used in React to make HTTP requests to a Spring Boot API?

A. fetch

B. Axios

C. jQuery

D. Lodash

2. Which HTTP method is typically used to create a new resource?

A. GET

B. POST

C. PUT

D. DELETE

3. What format is typically used to send data from React to a Spring Boot REST API?

A. XML

B. JSON

C. YAML

D. Plain Text

4. Which hook is commonly used in React to fetch data from an API when a component loads?

A. useState

B. useEffect

C. useContext

D. useRef

KEY TAKEAWAY Axios is the common React HTTP client; POST creates a resource; JSON is the standard payload format; useEffect is the hook that fetches data when a component loads.

KNOWLEDGE CHECK (2 OF 2)

One more multiple-choice question on status codes, plus a real code-recognition exercise from the curriculum worksheet.

5. Which status code indicates that a resource was created successfully?

A. 200 OK

B. 201 Created

C. 204 No Content

D. 400 Bad Request

CODE RECOGNITION -- WHAT DOES THIS PRINT?

```
const [data, setData] = useState(null);
const [loading, setLoading] = useState(true);

useEffect(() => {
  axios.get('/api/tasks')
    .then(res => { setData(res.data); setLoading(false); })
    .catch(err => {
      console.log('Error:', err.message);
      setLoading(false);
    });
}, []);

console.log(loading, data);
```

6. WHAT IS THE OUTPUT ON THE FIRST RENDER?

- **A. true, null**
- B. false, []
- C. false, undefined
- D. false, list of tasks (array)

KEY TAKEAWAY 201 Created signals a successful POST; the bare console.log runs during the first render, before the effect's fetch resolves -- so it prints the initial state: true, null.

"The API is the bridge between your UI and your business logic."

MODULE 35 COMPLETE

Frontend to API Integration

You can now connect a React frontend to a Spring Boot REST API with Axios and fetch, handle every response and failure class gracefully, and apply frontend security best practices.